

SARNY, Ukraine

WMO: 330880

Lat: 51.3128N

Lon: 26.6139E

Elev: 156

StdP: 99.46

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-15.1	-21.3	0.6	-18.2	-17.9	0.8	-14.3	11.4	1.1	9.8	0.6	1.0	340	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.9	20.5	29.1	19.6	27.5	18.8	21.7	28.3	20.7	26.7	19.8	25.2	2.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.6	24.0	18.7	13.8	23.1	17.9	13.1	22.3	63.8	28.4	60.4	26.8	57.3	25.1	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	6.8	5.5	-22.4	33.3	3.9	1.7	-25.2	34.5	-27.5	35.5	-29.7	36.4	-32.5	37.7		
(5)			-22.6	23.1	3.8	1.0	-25.4	23.8	-27.6	24.4	-29.8	25.0	-32.5	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-3.5	-2.1	2.2	9.2	14.5	17.9	19.9	18.9	13.8	8.1	3.1	-1.7	(6)
(7)		DBStd	9.38	5.84	5.70	4.69	4.28	4.15	3.43	3.23	3.27	3.84	4.19	4.63	5.40	(7)
(8)		HDD10.0	1738	417	339	242	64	9	0	0	0	9	87	209	361	(8)
(9)		HDD18.3	3788	676	572	499	275	131	50	21	32	142	316	456	619	(9)
(10)		CDD10.0	1167	0	0	2	40	149	237	306	277	124	29	3	0	(10)
(11)		CDD18.3	177	0	0	0	1	13	37	68	51	8	0	0	0	(11)
(12)		CDH23.3	1779	0	0	0	14	158	355	651	521	80	1	0	0	(12)
(13)		CDH26.7	518	0	0	0	3	33	87	209	170	16	0	0	0	(13)
(14)	Wind	WSAvg	1.9	2.2	2.2	2.3	2.0	1.9	1.8	1.6	1.5	1.7	1.9	2.0	2.0	(14)
(15)	Precipitation	PrecAvg	619	34	32	35	37	60	71	109	60	55	40	46	40	(15)
(16)		PrecMax	785	76	56	81	96	144	150	261	178	117	129	101	70	(16)
(17)		PrecMin	437	7	18	4	5	21	12	27	2	7	4	4	14	(17)
(18)		PrecStd	92	15	10	17	18	26	31	70	42	34	29	21	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	10.7	17.0	25.3	29.8	31.2	33.1	33.3	28.5	22.3	15.4	9.2	(19)
(20)			MCWB	5.8	7.6	9.8	15.4	17.9	20.2	21.5	20.7	18.4	15.3	11.8	7.8	(20)
(21)		2%	DB	6.2	7.8	13.9	21.5	27.3	29.2	30.9	30.7	25.6	19.3	12.8	7.4	(21)
(22)			MCWB	4.5	5.5	8.1	13.4	17.8	19.3	21.1	20.3	17.1	14.1	10.7	6.1	(22)
(23)		5%	DB	4.7	6.1	11.4	19.1	24.9	27.3	29.1	28.4	23.2	16.5	10.9	6.0	(23)
(24)			MCWB	3.4	4.3	7.2	12.1	16.7	18.6	20.1	19.4	16.2	12.8	9.5	5.0	(24)
(25)		10%	DB	3.1	4.4	9.0	16.9	22.6	25.3	27.2	26.2	20.8	14.6	9.3	4.5	(25)
(26)			MCWB	2.2	3.0	5.8	11.0	15.2	17.7	19.2	18.5	15.1	11.7	8.1	3.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	7.9	10.9	16.1	19.9	21.7	23.2	22.8	20.2	16.6	12.4	8.3	(27)
(28)			MCDWB	7.7	10.0	15.3	23.1	26.7	28.2	29.6	29.6	26.8	20.5	14.4	9.2	(28)
(29)		2%	WB	4.8	6.0	9.0	14.2	18.3	20.6	22.0	21.4	18.2	14.9	11.0	6.4	(29)
(30)			MCDWB	6.0	7.4	12.4	20.0	25.1	26.6	28.7	27.8	23.1	18.1	12.4	7.3	(30)
(31)		5%	WB	3.5	4.5	7.5	12.8	17.4	19.6	21.0	20.3	16.9	13.4	9.6	5.1	(31)
(32)			MCDWB	4.5	5.8	10.7	18.0	23.6	25.2	27.3	26.1	21.5	16.1	10.9	5.9	(32)
(33)		10%	WB	2.3	3.1	6.2	11.5	16.2	18.7	20.1	19.3	15.9	11.9	8.2	3.7	(33)
(34)			MCDWB	3.1	4.3	8.8	15.8	21.3	23.7	25.2	24.8	20.1	14.1	9.2	4.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	5.7	7.6	10.2	11.1	10.6	10.6	11.4	9.8	8.1	4.9	4.4	(35)
(36)			MCDBR	4.9	7.0	11.6	13.9	14.6	13.9	13.8	14.9	13.7	11.7	7.8	5.6	(36)
(37)			MCWBR	4.1	5.2	7.0	6.9	6.3	5.5	5.1	5.9	6.4	6.8	5.5	4.6	(37)
(38)		5% WB	MCDWB	4.4	6.4	10.3	12.6	12.9	11.8	12.1	12.4	11.6	10.2	6.8	5.3	(38)
(39)			MCWBR	4.0	5.1	6.6	6.7	6.2	5.3	5.0	5.6	6.1	6.5	5.1	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.342	0.376	0.427	0.413	0.404	0.434	0.433	0.401	0.361	0.344	0.307	(40)	
(41)		taud	2.211	2.163	2.196	2.153	2.267	2.316	2.241	2.238	2.280	2.369	2.304	2.258	(41)	
(42)		Ebn at Noon	682	759	809	808	842	853	817	795	777	741	640	623	(42)	
(43)		Edh at Noon	71	99	116	136	127	122	130	124	106	81	67	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.81	1.61	2.83	4.19	5.31	5.73	5.41	4.81	3.27	1.87	0.77	0.55	(44)	
(45)		RadStd	0.13	0.27	0.42	0.41	0.50	0.43	0.44	0.35	0.47	0.32	0.07	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.73	N/A	N/A	+0.97	+0.58	+0.47	-151	-232	+112	+40
(47) Regional (7 neighbors)		+0.64	N/A	N/A	+0.87	+0.61	+0.57	N/A	-197	+112	+44

Nomenclature: See separate page